III B.Tech – II Semester

Course Code	**HIGHWAY ENGINEERINGLAB**	L	T	P	C
23A01602P		0	0	3	1.5

Course Objectives:

The objectives of this course are to make the student to:

1. **Understand** the properties and behavior of aggregates and bitumen usedin highway construction.
2. **Perform** standard laboratory tests on aggregates and bitumen to evaluate their suitability for road construction.
3. **Analyze** the strength, durability, and performance characteristics of pavement materials.
4. **Assess** the quality and compliance of highway materials with standard specifications.
5. **Develop** hands-on skills for material testing and interpretation of test results.

Course Outcomes:

Upon successful completion of this course, students will be able to:

1. **Determine** the physical properties of coarse aggregates, such as specific gravity, water absorption, and shape characteristics.
2. **Evaluate** the mechanical properties of aggregates, including abrasion resistance, impact strength, and crushing value.
3. **Analyze** the physical and chemical properties of bituminous materials through standard tests.
4. **Perform** Marshall stability tests and assess the optimum binder content for bituminous mixes.
5. **Interpret** test results to assess the suitability of aggregates and bitumen for pavement construction.

CO – PO Articulation Matrix:-

Course Outcomes	**PO 1**	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**
CO -1	3	2	2	2	1	3	3	1	-	1	-	2	3	3
CO -2	3	3	3	2	2	2	2	1	-	-	-	1	3	3
CO -3	3	3	3	2	2	2	3	1	-	-	-	1	3	3
CO -4	3	3	3	2	2	2	3	1	-	-	-	1	3	3
CO -5	2	2	2	1	3	3	3	1	-	1	1	2	2	2

List of Experiments:-

I. TEST ON AGGREGATES

1. Specific Gravity Determination ofthe Coarse Aggregate Sample
2. Determination of Abrasion Value ofthe Coarse Aggregate Sample.
3. Determination of Impact Value of Coarse Aggregate
4. Determination of Elongation Index of Coarse Aggregate
5. Determination of Flakiness Index of Coarse Aggregate
1. Determination of Aggregate Crushing Value of Coarse Aggregate
2. Determination of Water Absorption Capacity ofthe Coarse Aggregate Sample.

II. TEST ON BITUMEN

1. Specific Gravity Determination ofThe Bitumen/Asphalt Sample.
2. Penetration Test OnBitumen.
3. Viscosity Determination of Bituminous Binder .
4. Determination of Softening Point of The Asphalt/Bitumen Sample
5. Determination of Ductility Value of The Bitumen Sample
6. Estimation of Loss of Bitumen On Heating
7. Bitumen Extraction Test

TEXT BOOKS:

1. Highway Material Testing Manual, Khanna, Justo and Veera Raghavan, Nemchand Brothers

REFRENCE BOOKS:

1. IS 383 :1993 “Specification for Coarse and Fine Aggregates From Natural Sources for Concrete”
2. 1S 1201 -1220 (1978) “Methods for testing tars and bituminous materials”
3. IRC SP 53 -2010 “Guidelines on use of modified bitumen”
4. MS-2 Manual for Marshalls Mix design 2002

Online Learning Resources:

https://ts-nitk.vlabs.ac.in/